Digital Inclusion and Circular Economy Behavior in Indian States: A Digital Circularity Index and Explainable XGBoost Analysis

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***Abstract*—India is simultaneously expanding digital infrastructure and pursuing circular economy goals, yet the relationship between these two agendas remains underexplored at the state level. This study integrates seven official government data sources to construct a Digital Circularity Index (DCI) for 16 Indian states. The analysis combines OLS regression, robustness testing, mediation analysis, threshold analysis, leak-free nested cross-validation, SHAP-based explainability, and K-Means clustering. Across methods, states with stronger digital access consistently show better circular economy outcomes. The association remains significant under non-parametric and permutation-based checks, while tele-density emerges as a stronger driver than digital payments. The results also reveal a substantial gap between digitally advanced and lagging states. These findings suggest that digital inclusion and sustainability policy should be treated as interconnected rather than separate agendas.**

***Index Terms*—Circular economy, digital circularity index (DCI), digital inclusion, extreme gradient boosting (XGBoost), Shapley additive explanations (SHAP)**

I. Introduction

Since 2015, Digital India has expanded broadband, mobile connectivity, and the Unified Payments Interface (UPI) across a country of 1.4 billion people. Monthly UPI transactions now exceed 14 billion [1], and national tele-density is above 85% [2]. At the same time, India generates roughly 62 million tonnes of waste annually and ranks among the world's top three e-waste producers [3]. NITI Aayog's circular-economy (CE) strategy now spans 11 priority sectors [4].

Yet the relationship between these agendas remains underexplored. Digital India research focuses mainly on growth, employment, and financial inclusion [5], while CE research stays centered on waste-management technology and EPR policy [6]. Very little state-level work asks whether stronger digital access is associated with better environmental performance.

A. What This Paper Does

This paper addresses that gap by constructing a Digital Circularity Index (DCI) from seven official data sources covering 16 Indian states. The analysis combines OLS, non-parametric tests, bootstrap resampling, seven LOO-CV models, nested-CV XGBoost with SHAP, and K-Means clustering. The main contributions are a state-level index linking digital access and CE behavior, cross-method validation of the relationship, and state-specific interpretation through clustering and explainability.

B. Research Questions and Hypotheses

The central question is straightforward: does digital inclusion predict circular economy performance across Indian states? We formalised this into three testable hypotheses:

- **H1:** States with higher Digital Access Scores will also have higher CE Behavior Scores.
- **H2:** UPI adoption mediates this relationship—i.e., digital payments are the channel through which digital access affects CE behavior.
- **H3:** There is a tele-density threshold beyond which CE behavior improves sharply.

II. Literature Survey

A. Digital Inclusion in India

Between 2015 and 2024, India pushed tele-density beyond 85% nationally, while some urban pockets crossed 130% [2]. Since 2016, UPI has grown into the world's largest real-time digital payments system [1]. But large regional disparities remain: Bihar's tele-density is around 55.80, while Kerala exceeds 120%. Most research in this area focuses on growth, access expansion, and financial inclusion [5], leaving the link between digital connectivity and sustainability outcomes relatively underexplored.

B. Circular Economy in India

The circular economy entered the policy mainstream through the Ellen MacArthur Foundation and was later distinguished more clearly from the broader idea of sustainability by Geissdoerfer *et al.* [6], [7]. In India, NITI Aayog translated that agenda into an 11-sector strategy [4], but implementation remains uneven. India generates roughly 1.71 million tonnes of e-waste annually, only part of which enters formal recycling channels [3], and state SDG scores still range from 57 to 79 [8].

C. Where Digital and CE Research Miss Each Other

Avgerou and Heeks argue that ICTs reshape development through multiple channels, including governance, markets, information flow, and institutional functioning [9], [10]. If that view holds, digital access should affect CE behavior through several mechanisms rather than one isolated pathway. Even so, state-level evidence connecting Digital India with circular-economy outcomes remains limited. This study addresses that gap.

TABLE I
DATA SOURCES AND VARIABLES

Source	Variable	Year
TRAI [2]	Tele-density (%)	2024
RBI [11]	Per Capita NSDP (INR)	2023–24
NITI Aayog [8]	SDG Index Score	2023–24
NPCI/ Dataful [1], [12]	UPI Per Capita Txns	2024–25
CPCB [3]	E-Waste Processed (T)	2021–22
Census [13]	Literacy Rate (%)	2011
Census [13]	Urban Population (%)	2011

III. METHODOLOGY

A. Research Design

This study uses a **cross-sectional** design covering 16 Indian states with complete observations across all seven government sources. We apply an **inner-join strategy**, favoring consistency and traceability over a larger sample.

B. Data Sources

After merging the source files, the final dataset contains **16 states** $\times$ **15 variables** (9 raw and 6 derived).

C. Composite Index Construction

1) Min-Max Normalization: Each raw variable x_i is rescaled to a $[0, 100]$ range:

$$\hat{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \times 100 \tag{1}$$

2) Digital Access Score (DAS):

$$DAS = 0.40 \cdot \hat{T}_d + 0.30 \cdot \hat{U}_{pc} + 0.30 \cdot \hat{N}_{pc} \tag{2}$$

where $\hat{T}_d$ denotes normalized tele-density, $\hat{U}_{pc}$ denotes normalized UPI transactions per capita, and $\hat{N}_{pc}$ denotes normalized per capita NSDP. Tele-density receives the largest weight because it captures underlying communications infrastructure.

3) CE Behavior Score (CEBS):

$$CEBS = 0.35 \cdot \hat{E}_{pc} + 0.40 \cdot S_{sdg} + 0.25 \cdot \hat{N}_{pc} \tag{3}$$

where $\hat{E}_{pc}$ is normalized e-waste processed per capita, S_{sdg} is the SDG Index Score, and $\hat{N}_{pc}$ is normalized per capita NSDP.

4) Digital Circularity Index (DCI):

$$DCI = \frac{DAS \times CEBS}{100} \tag{4}$$

This multiplicative form requires strength on both dimensions.

D. Statistical Analysis Methods

H1 is evaluated using an OLS regression of CEBS on DAS:

$$CEBS_i = \beta_0 + \beta_1 \cdot DAS_i + \varepsilon_i \tag{5}$$

Diagnostics include Breusch-Pagan, Durbin-Watson, and Cook's distance. H2 is tested with the Baron & Kenny [14] framework and the Sobel test, while H3 uses piecewise regression with grid-searched threshold estimation.

TABLE II
DESCRIPTIVE STATISTICS OF COMPOSITE INDICES (n = 16)

Statistic	DAS	CEBS	DCI
Mean	64.95	46.98	32.52
Std. Dev.	19.69	13.90	17.40
Min	30.27	27.23	8.24
Max	94.40	87.30	72.97
Median	67.02	47.78	32.42

E. Machine Learning Methods

1) Leave-One-Out Cross-Validation (LOO-CV): With only $n = 16$ observations, we use LOO-CV:

$$R^2_{LOO} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_{-i})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6}$$

Seven models are compared under this framework.

2) Nested LOO-CV with GridSearch: To avoid hyperparameter leakage, we also implement nested cross-validation with an outer LOO loop for evaluation and an inner 4-fold loop for tuning.

3) SHAP Explainability: Model explainability is handled using SHAP [15]:

$$f(x) = \phi_0 + \sum_{j=1}^{p} \phi_j(x) \tag{7}$$

where ϕ_0 is the baseline prediction and $\phi_j(x)$ is the contribution of feature j for instance x.

4) K-Means Clustering: Finally, K-Means clustering is used to identify state groupings, with the number of clusters selected by silhouette score.

IV. RESULTS

A. Descriptive Statistics

Table II summarizes DAS, CEBS, and DCI across the 16 states. Bihar records the lowest value on all three indices, while Haryana posts the highest DCI at 72.97 and Karnataka the highest DAS at 94.40.

Figs. 1 and 2 show the spread clearly. SDG Index, Per Capita NSDP, and Tele-density display the strongest associations with CE Behavior Score.

B. Primary Regression Analysis (H1)

The fitted OLS model is:

$$CEBS = 11.158 + 0.551 \cdot DAS \tag{8}$$

A one-point increase in DAS is associated with roughly a 0.55-point increase in CEBS. The model explains 61% of the variance, and the coefficient remains significant at $p < 0.001$, supporting H1.

A five-control specification raises R^2 to 0.763, but it also introduces clear multicollinearity: the VIF reaches 10.57 for DAS and 9.47 for NSDP. Because NSDP is built into both composite scores, the bivariate model in Eq. 8 remains the cleaner test of H1.

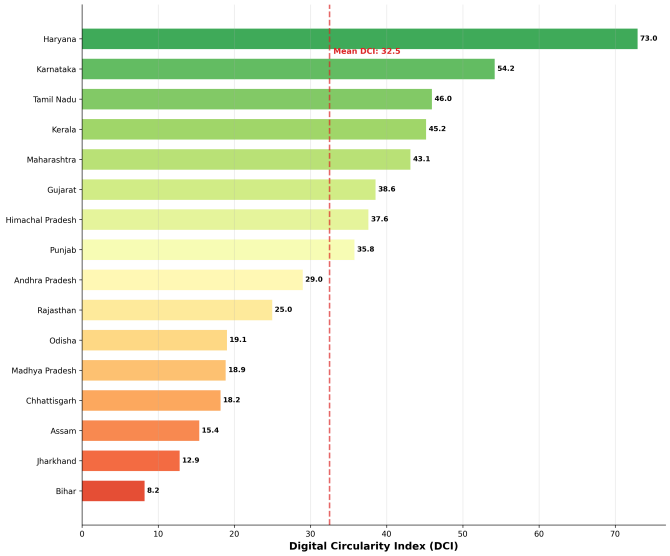


Fig. 1. Digital Circularity Index (DCI) ranking across 16 Indian states. Haryana leads (72.97), Bihar lags (8.24).

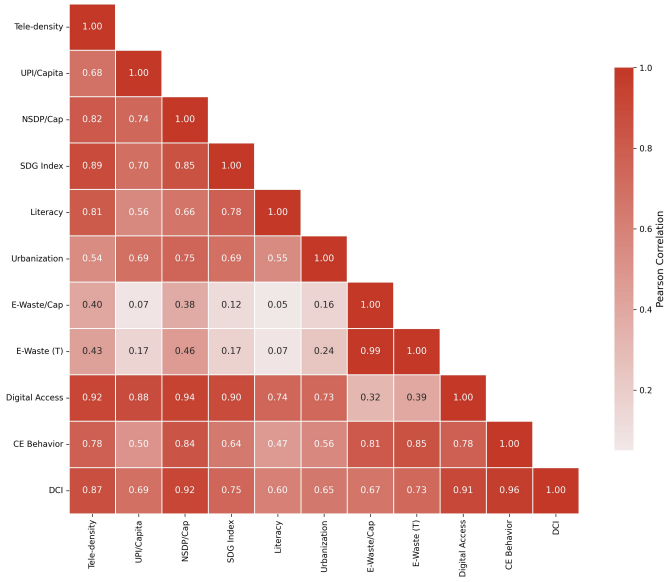


Fig. 2. Correlation heatmap of all numeric variables. Darker colors indicate stronger correlations.

C. Robustness Checks

Residual diagnostics showed non-normality in the OLS model and strong influence from Haryana (Cook's $D = 4.16$), so we tested the result under less favorable assumptions. Table IV shows that the association remains stable with HC3 standard errors, permutation testing, and rank-based statistics. It also survives alternative weighting schemes: six of seven variants produce significant Pearson correlations, and the only borderline UPI-heavy case still yields a strong Spearman correlation ($\rho = 0.882$, $p < 0.001$).

TABLE III
OLS Regression Results — Primary Model

Metric	Value
R^2	0.610
Adjusted R^2	0.583
β_1 (slope)	0.551
SE(β_1)	0.118
t-statistic	4.684
p-value	0.000352
F-statistic	21.94
95% CI for β_1	[0.299, 0.804]
Pearson r	**0.781**

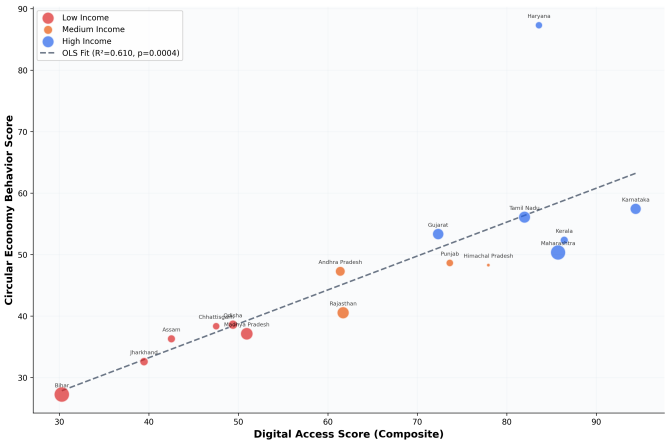


Fig. 3. Digital Access Score vs. CE Behavior Score ($r = 0.781$, $p < 0.001$). Each point represents one state.

TABLE IV
Robustness Diagnostics

Test	Value	Result
Spearman ρ	0.912	$p < 0.001$
Kendall τ	0.783	$p < 0.001$
Shapiro-Wilk	Skew = 2.72	Non-normal
Breusch-Pagan LM	1.539	$p = 0.215$
Durbin-Watson	2.025	No autocorr.
Cook's distance	$D_{\max} = 4.16$ (HR)	1 infl. pt.
Permutation p	0.0001	Significant
HC3 robust p	0.000011	Significant
Bootstrap CI (r)	[0.648, 0.979]	Excludes 0
Bootstrap CI (β_1)	[0.395, 0.836]	Excludes 0

D. Mediation Analysis (H2)

H2 tests whether UPI adoption mediates the relationship between digital access and CE behavior.

The evidence does not support mediation through UPI. The Sobel test is not significant ($z = -0.241$, $p = 0.810$), while the direct effect remains significant after controlling for UPI ($c' = 0.530$, $p = 0.004$). Two alternative pathways—Tele-density → NSDP → CEBS and Tele-density → SDG → CEBS—are also unsupported.

TABLE V
MEDIATION RESULTS

Path	β	p-value	Sig.?
c (total): $X \to Y$	0.505	0.0004	Yes
a: $X \to M$	1.499	0.003	Yes
b: $M \to Y \mid X$	−0.017	0.813	No
c' (direct): $X \to Y \mid M$	0.530	0.004	Yes
Indirect ($a \times b$)	−0.026	—	—
Sobel z	−0.241	0.810	No

TABLE VI
ML MODEL COMPARISON — LOO-CV RESULTS

Model	Digital-Only		Full Model	
	R^2	r	R^2	r
ElasticNet	**0.364**	0.603	0.534	0.731
Lasso	0.347	0.606	**0.568**	0.758
Ridge	0.338	0.597	0.536	0.743
RF	0.326	0.575	0.490	0.703
Lin. Reg.	0.301	0.585	0.442	0.713
SVR (RBF)	0.286	0.554	0.425	0.700
GB	0.161	0.542	0.445	0.687

E. Threshold Analysis (H3)

The estimated tele-density breakpoint is 118.77%:

- Below that threshold, the slope is steep and positive: $\beta_1 = 0.635$ ($p = 0.0001$).
- Above it, the slope flips negative: $\beta_2 = -8.738$ ($p = 0.047$)—but only two or three states sit in this range, so we would not read too much into the reversal.
- The piecewise model fits better than the linear one: $R^2 = 0.723$ vs. 0.610.

These results provide qualified support for H3. CE outcomes improve sharply up to roughly 119% tele-density, but only a few states lie above the breakpoint.

F. Machine Learning Validation (LOO-CV)

We also tested whether the signal survives out-of-sample prediction under LOO-CV.

Even with only the three digital-access variables, ElasticNet explains 36% of the out-of-sample variance in CE behavior ($p = 0.013$). The best full-model LOO-CV result (Lasso, $R^2 = 0.568$) also remains close to the OLS value of 0.610.

G. Nested LOO-CV (Leak-Free Evaluation)

To remove hyperparameter optimism from standard LOO-CV, we used a nested design with an outer LOO loop and an inner 4-fold tuning loop.

XGBoost performs best under this stricter setup, reaching $R^2 = 0.511$ ($p = 0.002$), while SVR falls below a mean-prediction baseline. In repeated 4-fold CV, Random Forest shows the tightest distribution, with Mean $R^2 = 0.442$ and Median $R^2 = 0.603$.

TABLE VII
NESTED LOO-CV RESULTS (LEAK-FREE)

Model	Nested R^2	RMSE	r
XGBoost	**0.511**	9.41	0.721
Random Forest	0.504	9.48	0.711
ElasticNet	0.471	9.78	0.723
Ridge	0.445	10.03	0.714
Lasso	0.443	10.05	0.713
SVR	−0.023	13.61	−0.593

TABLE VIII
SHAP GLOBAL FEATURE IMPORTANCE

Rank	Feature	Mean $\lvert\phi\rvert$
1	Tele-density (%)	**5.012**
2	Per Capita NSDP	4.883
3	Population (Mn)	1.714
4	Urban (%)	1.677
5	Literacy Rate (%)	0.998
6	UPI Per Capita	0.914

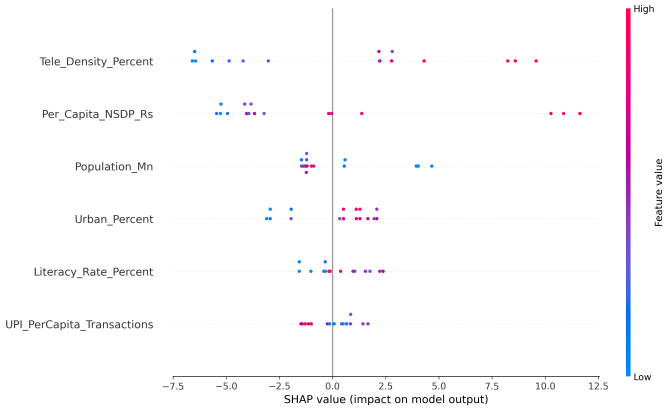


Fig. 4. SHAP beeswarm plot. Each dot represents one state for one feature. Color indicates feature value (red = high, blue = low). Tele-density and NSDP have the widest spread.

H. SHAP Explainability Analysis

SHAP values were computed for all six features across all 16 states.

Tele-density ranks first (mean $|\phi| = 5.01$), with Per Capita NSDP close behind at 4.88. UPI ranks last at 0.91. Figs. 4 and 5 show the same pattern at both global and state levels.

I. K-Means Clustering (State Typologies)

K-Means clustering produced a best-fit solution of $k = 2$ (Table IX). The two groups correspond to **Digital Leaders** and **Emerging States**.

The separation is substantial: average DCI differs by roughly 2.5× (46.7 vs. 18.3). PCA reinforces the same interpretation, with the first component explaining 62.3% of the variance.

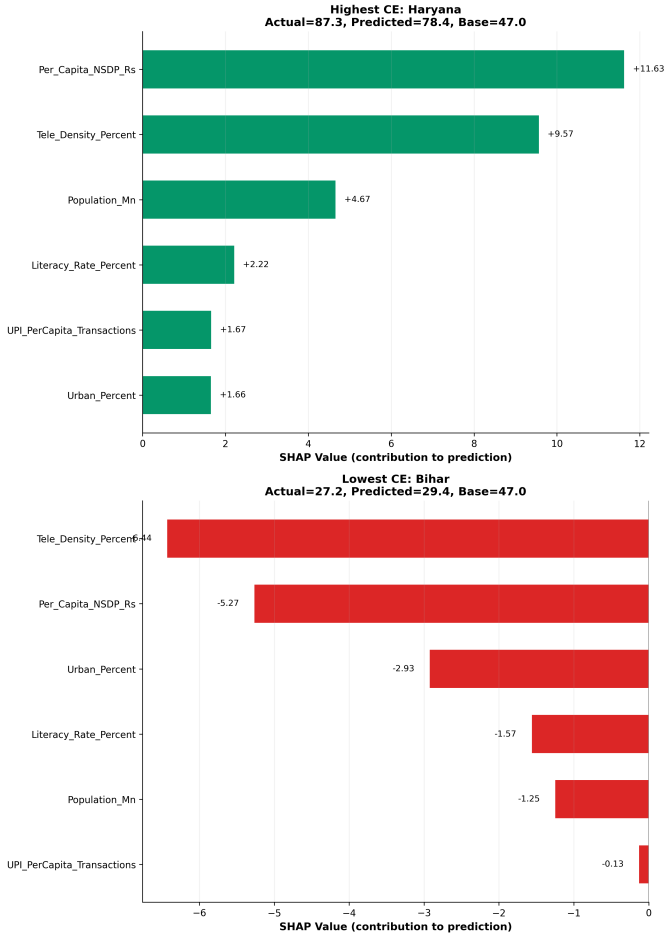


Fig. 5. SHAP waterfall explanations. Left: Haryana (highest CE)—high NSDP (+11.63) and tele-density (+9.57) push score up. Right: Bihar (lowest CE)—low tele-density (−6.44) and NSDP (−5.27) push score down.

TABLE IX
K-MEANS CLUSTER PROFILES

Cluster	n	**DAS**	**CEBS**	**DCI**
Digital Leaders	8	82.0	56.7	**46.7**
Emerging States	8	47.9	37.2	18.3
Gap	—	34.1	19.5	**28.4**

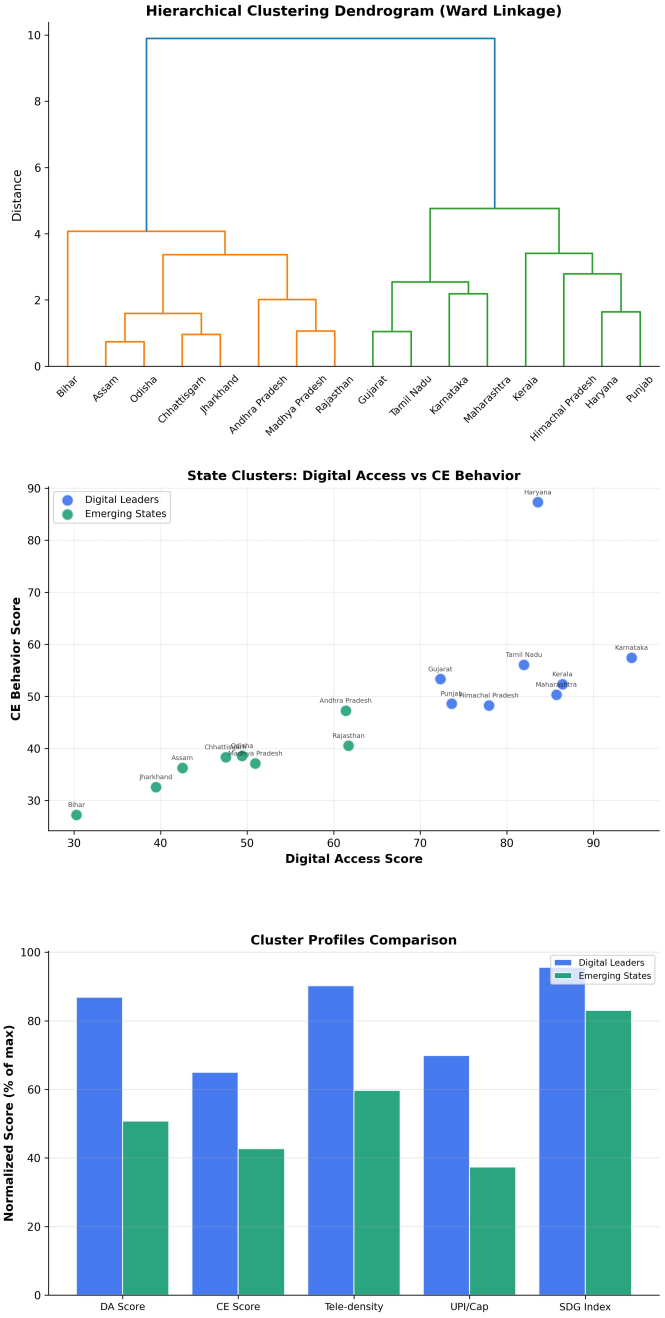


Fig. 6. K-Means clustering results with hierarchical dendrogram and cluster radar profiles. Two distinct state typologies emerge.

J. Extended Sample Validation

As an extended-sample check, we removed e-waste from CEBS and recalculated the analysis, increasing the sample to $n = 21$. The relationship became stronger rather than weaker: Pearson $r = 0.963$ ($p < 0.000001$), Spearman $\rho = 0.951$, and OLS $R^2 = 0.927$.

V. DISCUSSION

A. Do the Methods Agree?

Broadly, yes. The OLS model explains 61% of the variance, while nested-CV XGBoost explains 51.1%. The expected drop from in-sample to out-of-sample performance does not change the main point: the relationship remains positive and significant across parametric, non-parametric, and predictive methods. For a 16-state sample, that consistency is meaningful evidence against a simple overfitting story.

B. What Drives the Relationship?

The feature-importance results differ in emphasis but not in substance: Random Forest favors Per Capita NSDP, while SHAP favors tele-density. Both methods nonetheless identify tele-density and NSDP as the two main drivers. UPI is

the more unexpected result. It receives the smallest SHAP contribution, and the Sobel test is not significant ($z = -0.241$, $p = 0.810$), suggesting that digital access matters less through payments alone than through broader infrastructure and institutional capacity [9], [10].

C. The Two Indias in CE Performance

The clustering results sharpen the divide: one group is strong on both digital access and CE outcomes, while the other is weak on both. The DCI gap—46.7 versus 18.3—suggests not a mild gradient but two distinct development settings, with the digital divide carrying a clear sustainability dimension.

D. What This Means for Policy

1) Digital India and Swachh Bharat should be treated less as separate policy tracks and more as connected systems.
2) Connectivity investment in lagging states should be viewed not only as a telecom intervention but also as part of environmental capacity-building.
3) Because UPI is not the mediating channel, digital inclusion policy should extend beyond payments to reporting systems, e-governance, and digital literacy.
4) The two clusters argue against a single national playbook: Digital Leaders can move toward advanced monitoring, while Emerging States need basic infrastructure first.
5) SHAP highlights state-specific constraints, suggesting that uniform intervention design would miss important differences.

E. Limitations and Future Work

The study has clear limits. The final sample contains only 16 states because we retain only complete and directly comparable observations across all seven sources. That improves traceability but increases sensitivity to influential cases such as Haryana (Cook's $D = 4.16$). The bootstrap intervals and 21-state validation strengthen confidence in the result, but they do not remove the small-sample constraint.

The design is also cross-sectional, so the analysis supports association rather than causation. Stronger causal claims would require panel data and designs that can separate policy effects from background factors such as governance quality and long-run development differences.

Measurement constraints remain. Literacy and urbanisation still come from the 2011 Census, NSDP enters both DAS and CEBS, and state averages compress large within-state variation. Future work should therefore use district-level or panel data, expand the CE index with measures such as plastic recovery and EPR compliance, and test the framework in other countries with comparable administrative data.

VI. Conclusion

This study finds a consistent state-level association between digital access and circular-economy behavior in India. Across the 16 states retained in the final dataset, the pattern remains visible under both statistical and machine-learning approaches: the OLS model gives $r = 0.781$ ($p < 0.001$), while leak-free nested-CV XGBoost still reaches $R^2 = 0.511$.

Tele-density matters more than digital payments in explaining the observed pattern, and the clustering analysis reinforces the same message by separating states into two groups with a 2.5× DCI gap. Taken together, these results suggest that digital inclusion and circular-economy policy should be treated as connected rather than separate agendas.

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